

Validation of CCD Monitoring for GAST-D GBAS

Dong-Kyeong Lee⁽¹⁾, Jiyun Lee*

*Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology,
Daejeon, Republic of Korea
jiyunlee@kaist.ac.kr*

Introduction

Ground Based Augmentation System (GBAS) is able to provide Global Positioning System (GPS) signal corrections for users in the vicinity of the ground station to achieve high accuracy and precision mission operations. It is currently installed at several airports around the world and its numbers are steadily increasing. Although Instrument Landing System (ILS) is still being used as the system for precision approach operations, future implementation of GBAS is expected to share or take over the roles of ILS. GBAS has lower initial and maintenance costs than ILS as multiple receivers are not necessary, can provide auto landing for aircrafts in zero visibility environments, and its introduction will reduce the congested Very High Frequency Navigation (VHF NAV) band. However, to successfully implement GBAS, the system needs to satisfy the necessary aviation standards identified in International Civil Aviation Organization (ICAO) Standards and Recommended Practices (SARPs). Current GBAS standards only augment single frequency GNSS and supports landings for Category-I (CAT-I) minima, which is also known as GBAS Approach Service Type C (GAST-C). If GBAS is to be used for precision landing and approach, it needs to satisfy the Category-II/III (CAT-II/III) minima, which is known as GAST-D. Although various ground and airborne monitors for GBAS need to be considered for GAST-D, this paper will focus on the validation of the CCD monitor for GAST-D.

CCD Monitor for GAST-D GBAS

One of the challenging issues for GAST-D GBAS service is the mitigation of the ionospheric threat. There are two mitigation approaches for the characterization of the ionosphere threat space:

Ground Subsystem Mitigation Approach

- Code Carrier Divergence (CCD) Monitor
- Ionosphere Gradient Monitor

Airborne Subsystem Mitigation Approach

- Range Domain DSIGMA Monitor
- Code Carrier Divergence (CCD) Monitor
- Full DRAIM Monitor

For the validation of CCD monitoring for GAST-C, only the ground subsystem was considered. For this research, as we are dealing with GAST-D, both the ground and airborne CCD monitors are used.

The algorithms used for the CCD monitor has been adopted from works by Simili [1].

Validation Methodology

A. GAST-D GBAS Operation Requirements

The minimum accuracy for lateral and vertical errors of CAT III system are specified in RTCA DO-245A, Minimum Aviation System Performance Standards for Local Area Augmentation System (LAAS) [2].

B. CCD Monitor Validation Environment

The ionosphere anomaly model we will be considering for the validation is the wedge model. The form of the wedge model is identical to the model developed by Stanford University for the FAA [3, 4]. The model is defined as a linear increase in ionosphere delay over a particular distance, or width. The wedge model can be divided into two cases, where the ionospheric front is slower than or faster than the aircraft. The slower case is called the slow front, while the faster case is called the fast front.

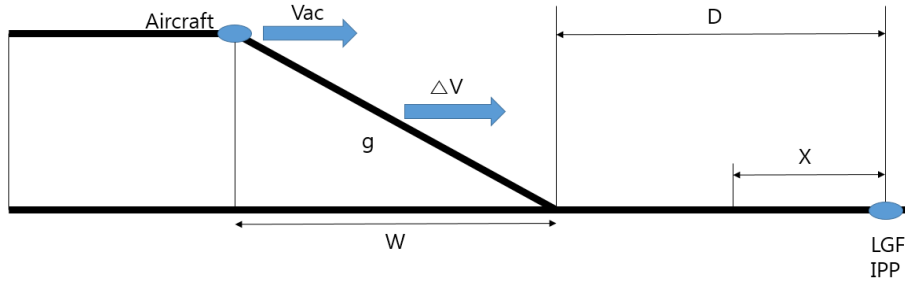


Figure 1 Slow Front Wedge Model

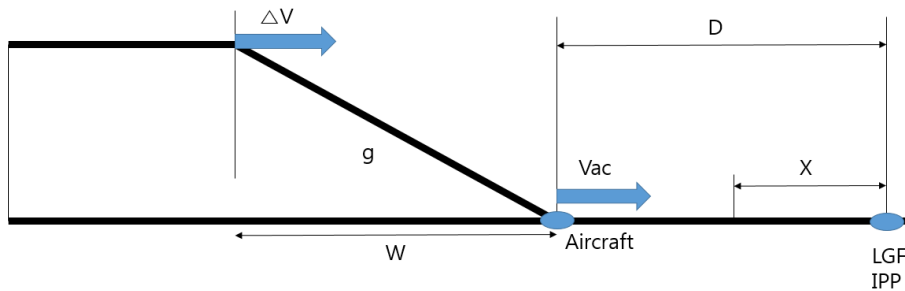


Figure 2 Fast Front Wedge Model

V_{ac} is the velocity of the aircraft, Δv is the gradient front speed relative to the LAAS Ground Facility (LGF) Ionospheric Pierce Point (IPP), X is the decision height, D is the initial distance between the gradient front and the LGF IPP, g is the gradient slope, and W is the width of the gradient region.

The values for the above variables are determined based on the SARPs GAST-D section D.7.5.6.1.7.1 [5] and Navigation System Panel (NSP) [6]

Threshold CCD_{LGF}	0.043 m/s
Threshold CCD_{AC}	0.052 m/s
CCD Time Constant	10 seconds
CSC Time Constant	100 seconds
Gradient Slope	0 – 500 mm/km
W	25 – 200 km
X	6 km
D	0 – 40 km
V_{AC}	70m/s

Results and Discussion

The Figure 3 shows the undetected range error due to ionospheric anomaly when $\Delta v = 60\text{m/s}$, $g = 400\text{mm/km}$. The green line indicates the range error experienced by the aircraft, while the black line indicates the range error experienced at the LGF IPP. The blue line shows the total undetected range error when CCD monitor is unavailable, and the red line indicates the undetected errors when CCD monitor is used. From the figure, it is evident that the monitor lowers the total undetected range error.

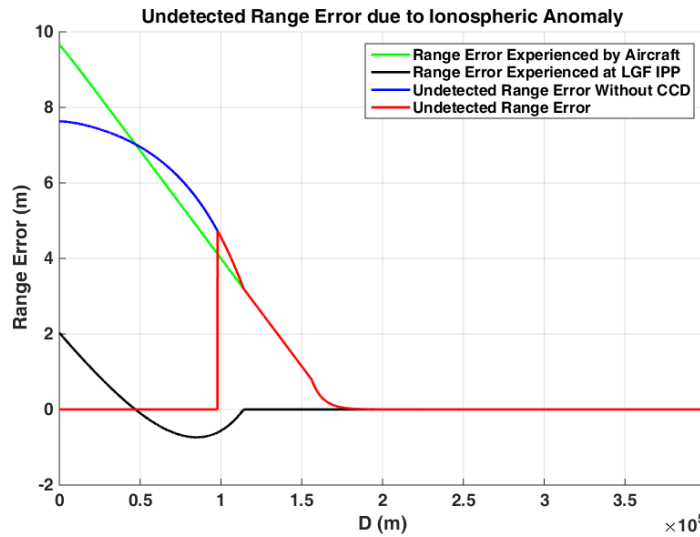


Figure 3 Undetected Range Error due to Ionospheric Anomaly

The Figure 4 shows the threat space for the maximum undetected range errors for different gradient slopes and relative gradient front speeds. When the gradient increase, the influence of the anomaly increases, so the undetected errors increase. For the relative gradient front speeds, for the slow front, the increase in the speed of the front decreases the range error, while for the fast front, the increase in the speed of the front increases the range error. Therefore, when the gradient front speed is closest to that of the velocity of the aircraft, the undetected error is minimal.

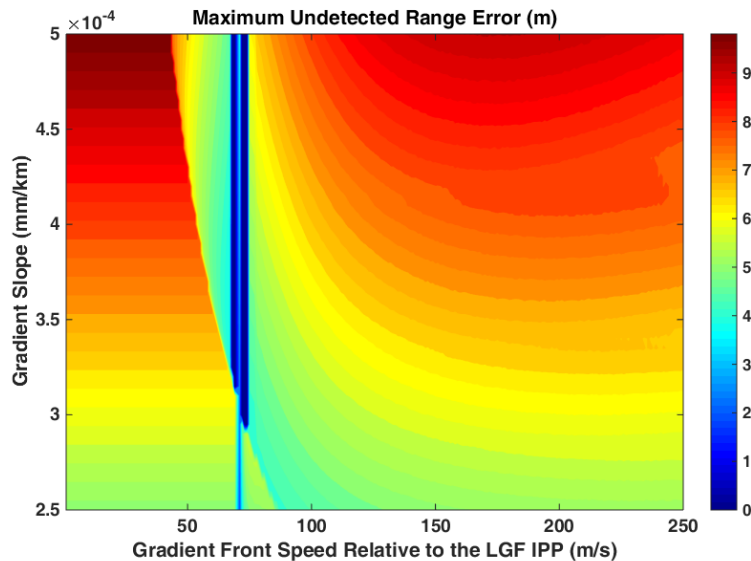


Figure 4 Threat Space Showing the Maximum Undetected Range Error

A new GAST-D CCD monitor is designed and demonstrated under the original GAST-C framework. Compared to GAST-C, due to the additional airborne CCD monitor available on the aircraft, the maximum range error due to misdetection has decreased. However, as the maximum range error is far greater than 2.75m which is the minimum allowable undetected range error for GAST-D monitors, CCD alone cannot satisfy the requirements. Therefore, more analysis with additional monitors are required and further improvements may be needed for the current CCD monitor to provide GAST-D service.

Acknowledgement

This research was supported by the Space Core Technology Development Program of the National Research Foundation (NRF) funded by the Ministry of Science, ICT & Future Planning (NRF-2014M1A3A3A02034937)

References

- [1] Dwarakanath V. Simili and Boris Pervan, “Code-Carrier Divergence Monitoring for the GPS Local Area Augmentation System,” Position, Location, and Navigation Symposium, 2006 IEEE/ION
- [2] “Minimum Aviation System Performance Standards for Local Area Augmentation System,” RTCA, DO-245A, Dec. 9, 2004
- [3] Luo, M. et al., “Ionosphere Spatial Gradient Threat for LAAS: Mitigation and Tolerable Threat Space,” Proceedings of the ION National Technical Meeting, Jan. 2004
- [4] Luo, M. et al., “LAAS Study of Slow-Moving Ionosphere Anomalies and Their Potential Impacts,” Proceedings of the Institute of Navigation, GNSS Conference, Sept. 2005
- [5] “Standards and Recommended Practices,” International Civil Aviation Organization
- [6] Working Paper, “GAST D Operational Validation / Review of Validation Material for Ionosphere: Iono Gradient Mitigation – Work Progress and Work Plan,” Navigation Systems Panel First Meeting, 1 to 11 December 2015, Presented by Tim Murphy, Prepared by IGM Ad-hoc